

Through this postdoc fellowship, I plan to use culturally-sensitive lenses to challenge the coloniality of AI through 2 projects (1) expanding the definition of AI safety, privacy, and ethics by pluriversal experiences of impacts on people across the world and (2) highlighting resistance social movements against AI data centers.

1 Project 1

I want to create a framework to decolonize AI safety research that de-centers Western neoliberal understandings of safety, ethics, responsibility, and privacy. The framework will evaluate epistemic & physical violences witnessed by people involved in AIs' development and operations. I will contrast how Big Tech AI models and infrastructure disproportionately impacts people in and outside the imperial core (generally, WEIRD populations) and the periphery (generally, the majority world) and people who sit at their boundaries (Indigenous peoples in Western countries). This framework aims to unearth the extractive logics embedded in every layer of BigTech AI lifecycle, from data collection to model training, from extraction of minerals that form the physical AI infrastructure to privacy policies of the AI interfaces used by consumers. The framework will provide opportunities that shape AI safety that incorporates sustainability and sovereignty.

Scholars have noted that computing research and interventions reflect an imperial impulse [1–3]. However, without engaging with theories of change that center decoloniality, sovereignty, and liberation from the Majority World [4–6], AI ethics and safety research may result in neoliberal siloed interventions that fail to address the privacy violations of peoples harmed by the enormous power of the Big Tech AI enterprise. For example, trust & safety teams in Big Tech companies often tout their products as private, ethical, carbon-neutral, whereas, the same companies actively exploit miners who extract rare earth minerals in Congo and assemble their products in India. BigTech companies create their own rules for trust, safety, and ethics and show meaningless numbers that they abide by it. On a similar vein, what does privacy mean for exploited factory workers in India who watch abusive content or wear cameras to train AI robots that will soon replace them? Similarly, ChatGPT promises privacy settings that ensure users' data is not used for model training, yet also, legally allows abusers to generate fake nude images of their victims without consent. A narrow definition of ethics, privacy, & safety only brings surface level interventions that rarely addresses the underlying structural violence and prevents communities from achieving long-term healing and liberation.

Methods. My computing and critical theory expertise allows me to view technologies not just as a black box, but as a combination of abstract social and technical components. I study impacts of violence through personal technologies (e.g., mobile accounts, AirTags) [Late-Breaking NDSS 2024, Late-Breaking CHI 2024, USENIX 2024, CCS 2025, In-review USENIX 2027, CSCW 2025, CHI 2026, In-review CSCW 2027] and from physical infrastructures (e.g., AI data centers, AMBER alerts) [In-review CSCW 2027]. I will surface the underlying neoliberalism and coloniality in AI safety research across security & privacy, HCI, ethics, and critical computing venues. To do this, I will build on my workshop that calls for extending the definition of violence that transcends disciplinary boundaries [In-review Workshop DIS 2026] and situating resistance movements that counter the accelerated proliferation of AI use in planetary crises [In-review Workshop CHI 2027]. I will generate keywords of concepts that capture the power differentials among people involved in the development of an AI model using the 'Calculating Empires' framework [18]. Then, I will use automated scraping methods [In-review CSCW 2027] and snowball sampling to collect a wide collection of relevant research articles, news reports, and podcasts that formally or informally define ethics and privacy in global contexts. I will categorize these definitions into mode of harms, the actors involved (including people and the land), their impacts on actors, and the mitigation strategies to negotiate and address ethical and privacy concerns. Next, to ensure pluraliversality in the definitions and by situating them in place and space, I will interview key stakeholders embedded in a variety of aspects of the AI lifecycle. For example, organizers fighting AI data centers, survivors of gender-based violence who use AI for support, immigrants who face social media surveillance, or, researchers that work on AI safety research. The interview will utilize the Meta-Relationality protocol [19] to surface a diverse set of orientations around affect (feelings about different kinds of AI?), mitigation strategy (what theory of change informs their response to AI?), time (how does the time of shape their stance?), and agency (what kind of leverage is possible and required?). I utilized this framework to conduct a workshop with top CS and Math researchers [Workshop HLF 2026]. I sensed rising tensions and uncertainty among scholars in the midst of planetary crises and scientific acceleration and corporatization and lack of perceived agency to shape the future of AI.

Findings and Impact. Findings from the the study will result in a composite set of pluriversal contestations that show the agreements and contradictions across value-systems, social positions, and proximity to AI knowl-

edge creation and usage. These contestations are necessary to inform the privacy interventions so that they center agency and liberation from AI-mediated violence: whether that may be refusal on ethical grounds, regulatory action or guardrails, resistance movement, decentralized alternatives, sabotage from within academia or corporations, or, eco-social symbiotic technologies. This work re-imagines how sovereign and culturally-sensitive, sustainable, sovereign, and ethical AI can manifest and benefit the communities.

2 Project 2

Goal. The project aims to critically examine how settler-colonial extractive logics embedded within the proliferation of AI and data centers facilitate violence against rural and First Nations in Canada. Crucially, by centering the communities' refusal of AI data centers through social movements, we envision how sovereign and culturally-sensitive, sustainable, ethical, community-based and participatory AI can manifest and benefit the underserved communities.

Under the false promises of jobs, economic development, BigTech companies have accelerated constructions data centers targeting lands in rural, Indigenous and Global South places. Contrary to the foundational imaginary of the internet, there is an enormous capitalist push from BigTech and states towards AI inevitability, unlimited funding based on 'hype' and automation of human labor in essential sectors. Powering this imperial frontier are Data Centers, who thrive on natural resources like land water, electricity. However, such extractive proliferation facilitates violence that disenfranchise rural and Indigenous reservations; massive wildfires, job losses due to automation, ecological extraction, ecocide, climate catastrophe, inflated bills of rural and urban citizens [21–24]. Scholars question this imperial and settler-colonial logics that extract value from land water, electricity, labor from rural, Indigenous places and critical minerals from Global South. For e.g. Edwards [22] offer stories that run counter to Big Tech's performative neoliberal narrative of a clean and connective space that "*brings the world closer together*". Facebook manipulated the governments through 30-year real estate deals, temporary construction jobs, and billion-dollar tax breaks to steal cooled water pumped from desert aquifers in New Mexico, a state already affected by wildfires and droughts.

On the contrary, through embodied relational accountability, Indigenous values equate violence against land as violence against people. On one hand AI facilitates gender-based violence through deepfakes, non-consensual nude imagery [25], on the other, data centers complement the extractive mining and gas pipeline industries by creating temporary residential 'man camps' that mediate domestic violence and human/sex trafficking [In-review CSCW 2027]. This friction gives rise to an incompatible value-system where Indigenous communities treat land as a living relative in relational harmony; whereas capitalist modernity treats land and minerals as a resource to be extracted to satisfy individual freedom.

Methods. My prior work shows how Native communities resist structural violence and genocide facilitated by institutional technologies such as federal databases, social media algorithms, surveillance & policing [In-review CSCW 2027]. Disenfranchised by the AI frontier, Indigenous and rural peoples lead opposition movements refuse the privatization of land and extraction [26–28]. Insurgent movements led by Indigenous land defenders have resisted their land from settlers, companies, states and institutions through time immemorial. Similarly, climate agency coalitions, rural worker unions, and activists have organized within their communities to build resistance outside corporate supply chains, reclaiming technologies along the way. For e.g., activists use video-conferencing technologies to organize protests, community knowledge-sharing workshops, fair land-use policies, and pool resources such as toolkits, talking points, videos to share on social media websites [26]. My goal is to give voice to communities by answering: How proliferation of AI and data centers cause violence in rural and Indigenous communities of Canada? By centering the social movements that refuse violence and extraction, I will answer How do communities utilize socio-technical spaces to resist the AI data centers and seek safety? What challenges do they face? and How can we address the challenges and support their activism? Therefore, the work will generate a action-oriented scholarship towards alternative definitions and design of safety, ethical and responsible AI.

Findings and Impact. This work will result in metrics and theories that inform design of resilient and non-extractive infrastructure that replaces violence with collective healing by acknowledging Indigenous value-systems and sovereignty.

My Publications

- [Late-Breaking NDSS 2024] **Naman Gupta**, Rose Ceccio, Majed Almansoori, and Rahul Chatterjee. “Analyzing the Patterns and Behavior of Users When Detecting and Preventing Tech-enabled Stalking”. In: 2023. URL: <https://doi.org/10/gr8rnx>.
- [Late-Breaking CHI 2024] **Naman Gupta**, Sanchari Das, Kate Walsh, and Rahul Chatterjee. “A Critical Analysis of the Prevalence of Technology-Facilitated Abuse in US College Students”. In: *Extended Abstracts of the 2024 CHI Conference on Human Factors in Computing Systems*. 2024. URL: <https://doi.org/10.1145/3613905.3652036>.
- [USENIX 2024] **Naman Gupta**, Kate Walsh, Sanchari Das, and Rahul Chatterjee. ““I really just leaned on my community for support”: Barriers, Challenges, and Coping Mechanisms Used by Survivors of {Technology-Facilitated} Abuse to Seek Social Support”. In: 2024. URL: <https://dl.acm.org/doi/10.5555/3698900.3699179>.
- [CCS 2025] Julia Nonnenkamp, **Naman Gupta**, Abhimanyu Dev Gupta, and Rahul Chatterjee. “Hidden in Plain Bytes: Investigating Interpersonal Account Compromise with Data Exports”. In: *2025 ACM SIGSAC Conference on Computer and Communications Security*. 2025. URL: <https://doi.org/10.1145/3719027.3765147>.
- [In-review USENIX 2027] Julia Nonnenkamp, **Naman Gupta**, Angela Wang, Sophie Stephenson, and Rahul Chatterjee. “LEStrADE: Investigating Account Compromise using Data Exports for Survivors of Technology-Facilitated Abuse”. In: *review at USENIX Security*. 2026. URL: <https://naman.github.io/assets/files/papers/lestrade.pdf>.
- [CSCW 2025] Sophie Stephenson, **Naman Gupta**, Akhil Polamarasetty, Kyle Huang, David Youssef, Kayleigh Cowan, and Rahul Chatterjee. “Legal Evidence of Technology-Facilitated Abuse in Wisconsin: Surfacing Barriers Within and Beyond the Courtroom.” In: *Computer Supported Cooperative Work and Social Computing* 🏆 **Honorable Mention (top 5%)**. 2025. URL: <https://doi.org/10.1145/3757622>.
- [CHI 2026] Sophie Stephenson, **Naman Gupta**, Kyle Huang, David Youssef, Kayleigh Cowan, and Rahul Chatterjee. “Trauma-Informed Digital Evidence Collection: A Design Inquiry into Evidence Practices for Technology-Facilitated Abuse in Intimate Partner Violence”. In: *2026 CHI Conference on Human Factors in Computing Systems* 🏆 **Best Paper (top 1%)**. 2026. URL: <https://doi.org/10.1145/3772318.3790296>.
- [In-review CSCW 2027] **Naman Gupta**, Connie W. Chau, Sophie Stephenson, Rahul Chatterjee, and Kate Walsh. “Lessons from Trauma-Informed Training on Technology-Facilitated Abuse for Gender-Based Violence Advocates”. In: *review at CSCW (2026)*. URL: <https://arxiv.org/abs/2610.02688>.
- [In-review CSCW 2027] **Naman Gupta**, Sophie Stephenson, Chung Chi Yeung, Wei Ting Wu, Jeneile Luebke, Kate Walsh, and Rahul Chatterjee. ““Lighting The Way For Those Not Here”: How Technology Researchers Can Help Fight the Missing and Murdered Indigenous Relatives (MMIR) Crisis”. In: *review at CSCW 2027*. 2025. URL: <https://doi.org/10.48550/arXiv.2601.17966>.
- [In-review Workshop DIS 2026] Jasmine C Foriest, Alicia DeVrio, Li Qiwei, Nimra Ahmed, **Naman Gupta**, Jensine Raihan, Franchesca Spektor, and Justine Zhang. “Violence as a Boundary Object: Language, Power, and Design Justice”. In: *Review at ACM DIS Designing Interactive Systems*. 2026.
- [In-review Workshop CHI 2027] **Naman Gupta**, Freddy Reiber, Jasmine C. Foriest, Ailsa Robertson, Jordan Taylor, Palak Dudani, Ghadeer Awwad, Shreya Chowdhary, David Gamba,

Eugene Kim, and Katie Seaborn. "Counter-Computing: Digital Resistances and Global Solidarities". In: *Review at ACM CHI Conference on Human Factors in Computing Systems*. 2027.

[Workshop HLF 2026]

Naman Gupta. "Polycrisis in the Age of AI and Scientific Acceleration". In: 2026.

References

- [1] Paul Dourish and Scott D. Mainwaring. "Ubicomp's colonial impulse". In: *Proceedings of the 2012 ACM Conference on Ubiquitous Computing*. UbiComp '12. Association for Computing Machinery, 2012.
- [2] Syed Mustafa Ali. "A Brief Introduction to Decolonial Computing". In: *XRDS* (2016).
- [3] Jensine Raihan. "The Colonial Legacy of the Human in Human-Computer Interaction: Implications for Critical Design Approaches". In: *Proc. Sixth Decenn. Aarhus Conf. Comput. X Crisis*. AAR '25. Association for Computing Machinery, 2025.
- [4] Arturo Escobar. *Designs for the Pluriverse: Radical Interdependence, Autonomy, and the Making of Worlds*. Duke University Press, 2018.
- [5] Morgan Ndlovu. "Decoloniality Is Agency". In: (2025).
- [6] Paola Ricaurte. "Ethics for the Majority World: AI and the Question of Violence at Scale". In: *Media Cult. Soc.* (2022).
- [18] Kate Crawford and Vladan Joler. *Calculating Empires: A Genealogy of Technology and Power since 1500*. 2023.
- [19] Meta-Relationality Institute and Gesturing Towards Decolonial Futures Collective. *Clearing the Field: A Relational Protocol for Difficult Conversations about AI*. 2025.
- [21] Mel Hogan. "Big Data Ecologies". In: *ephemera: theory & politics in organization* (2018).
- [22] Dustin W. Edwards. "Digital rhetoric on a damaged planet: storying digital damage as inventive response to the anthropocene". In: *Rhetoric Review* (2020).
- [23] Dustin Edwards, Zane Griffin Talley Cooper, and Mél Hogan. "The making of critical data center studies". In: *Convergence* (2025).
- [24] Steven Gonzalez Monserrate. *The staggering ecological impacts of computation and the cloud*. 2022.
- [25] UN Women. *AI-powered online abuse: how AI is amplifying violence against women and what can stop it*. 2025.
- [26] Maria Renée. *Not one drop: how an arizona community came together to fight a data center*. 2026.
- [27] Honor The Earth. *No Data Centers on Native Land*. 2025.
- [28] Niamh Rowe. "US farmers are rejecting multimillion-dollar datacenter bids for their land: 'I'm not for sale'". In: *The Guardian* (2026).